

Pure Water San Diego: Advancing Sustainable Urban Water Management through Potable Reuse

Introduction

In the face of increasing water scarcity, climate change, and rapidly growing urban populations, cities around the world are being forced to reevaluate traditional approaches to water management. San Diego, California, offers a compelling example of forward-thinking water policy through its ambitious Pure Water San Diego program. This multi-billion-dollar potable reuse initiative is designed to provide a locally controlled, sustainable water supply by recycling wastewater into high-quality drinking water (*City of San Diego official website*, n.d.b). By 2035, the program is projected to meet nearly one-half of the city's total water demand, significantly reducing dependence on imported water from the Colorado River and Northern California, sources that are increasingly vulnerable due to drought, environmental degradation, and interstate competition (*Colorado region conditions*, n.d.).

Pure Water San Diego employs advanced water purification technologies such as membrane filtration, reverse osmosis, and ultraviolet light with advanced oxidation to ensure safety and quality (*City of San Diego official website*, n.d.b). Beyond the engineering, however, the project represents a profound shift in how communities perceive and interact with their water resources. This paper explores the development, goals, and phased implementation of Pure Water San Diego, evaluating the program within the broader context of the region's hydrological challenges, legal constraints, and public engagement efforts.

In doing so, it considers how the program contributes to long-term urban water resilience, environmental sustainability, and public trust. It also investigates how San Diego's experience may inform other urban centers seeking to integrate potable reuse into their water portfolios. Ultimately, this case study offers insight into the complex interplay between infrastructure, policy, science, and community in shaping the future of urban water systems in an era of uncertainty.

Background

San Diego faces ongoing and complex water resource challenges stemming from its semi-arid climate, recurrent drought conditions, and significant dependence on imported water (*Colorado region conditions*, n.d.). Historically, over 80% of the city's water supply has been imported from distant and increasingly stressed sources such as the Colorado River and the California State Water Project (*City of San Diego official website*, n.d.b). These external supplies are becoming less reliable due to a convergence of factors: prolonged drought intensified by climate change, degradation of aquatic ecosystems, and legal and political tensions among water users in California and neighboring states (*Colorado region conditions*, n.d.). This dependence on external water not only places San Diego at risk of physical water shortages but also exposes the city to rising water costs and supply volatility as upstream users and climate shifts further constrain availability.

Compounding these supply-side concerns is the issue of growing demand. San Diego's population now exceeds 3 million residents, and the wider metropolitan area continues to expand

(*San Diego metro area population 1950-2025*, n.d.). As the population grows, so too does pressure on the region's limited local water resources, which are insufficient to meet demand without substantial reliance on imported supplies. This situation underscores the urgent need for more resilient, local, and sustainable water strategies to secure the city's long-term water future.

The governance of San Diego's water system is complex, involving multiple agencies and overlapping jurisdictions (*SDCWA*, 2025). At the local level, the San Diego Public Utilities Department oversees daily operations, maintenance, and delivery of water services (*SDCWA*, 2025). State-level oversight is provided by the California State Water Resources Control Board, which sets water quality standards and enforces statewide regulations (*SDCWA*, 2025, Environmental Protection Agency, 2025a, Environmental Protection Agency, 2025b). Regional coordination is facilitated by the San Diego County Water Authority, which procures and distributes water for multiple jurisdictions in the area (*SDCWA*, 2025, Environmental Protection Agency, 2025b). These entities must work together to balance sometimes competing goals, including ensuring supply reliability, meeting conservation and regulatory mandates, and protecting environmental resources.

Key stakeholders influencing water policy include city and regional planners, environmental advocacy organizations, scientists, engineers, public health officials, and residents (*City of San Diego official website*, 2024). Their interaction has evolved over time, particularly since the 1990s when a proposed water reuse project faced widespread opposition (Schwartz, J, 2015). The project was dismissed in part due to negative public sentiment and media-driven

stigma around potable reuse, commonly dubbed “toilet to tap” (Schwartz, J, 2015). Today, a growing understanding of water scarcity and the scientific reliability of modern treatment technologies has helped shift public opinion. These dynamics underscore the importance of adaptive, inclusive, and transparent governance in navigating San Diego’s water future.

Description and Purpose

Pure Water San Diego is an ambitious, multi-phase potable reuse program designed to transform the city’s approach to water sustainability (*City of San Diego official website*, n.d.a, *City of San Diego official website*, n.d.b). Once fully implemented by 2035, the initiative aims to produce 83 million gallons of purified drinking water per day, enough to meet nearly one-half of San Diego’s total drinking water demand (*City of San Diego official website*, n.d.a). This bold goal represents a major step toward water independence in a region historically reliant on importing over 80% of its supply from distant and increasingly unreliable sources, such as the Colorado River and the California State Water Project (*City of San Diego official website*, n.d.b). At the heart of the project is a multi-barrier advanced treatment process that includes microfiltration, reverse osmosis, and advanced oxidation using ultraviolet light and hydrogen peroxide. This robust treatment train ensures that the final product not only meets but exceeds all federal and state drinking water standards, offering a safe and high-quality source of water for residents (Environmental Protection Agency, 2024).

The core objective of Pure Water San Diego is to enhance the city’s long-term water supply reliability and resilience, particularly in the face of growing climate pressures, drought

conditions, and population growth (*City of San Diego official website*, n.d.b). Beyond water supply security, the program has additional environmental and policy benefits. It significantly reduces the volume of treated wastewater discharged into the Pacific Ocean, helping to improve coastal water quality and reduce the ecological footprint of the city's wastewater system (*City of San Diego official website*, n.d.a). The initiative also serves as a model for sustainable, circular water management in urban settings.

Despite these considerable benefits, the project faces notable challenges. The capital investment required for infrastructure, treatment facilities, and distribution systems is substantial, totaling several billion dollars (*City of San Diego official website*, n.d.b). Operational costs, especially energy requirements for advanced treatment processes, are also high (*City of San Diego official website*, n.d.b). Public perception remains a persistent issue, while acceptance of potable reuse has grown, concerns linger among some residents about the safety and aesthetics of drinking recycled water. Additionally, questions of equity emerge regarding how the program engages marginalized communities, both in terms of public outreach and the distribution of benefits and costs. Ratepayers citywide will ultimately fund the system, and it is critical to ensure that no group is disproportionately burdened or left out of decision-making processes. Overall, Pure Water San Diego represents a forward-thinking but complex solution to a deeply rooted regional challenge.

Decision-making

Pure Water San Diego represents the culmination of decades of policy evolution and shifting public attitudes toward water reuse. Its roots can be traced back to the 1990s, when the city first proposed a similar wastewater recycling initiative (Schwartz, J, 2015). At that time, however, the project collapsed under the weight of public opposition and damaging media narratives that dubbed it the “toilet-to-tap” plan (Schwartz, J, 2015). These perceptions, combined with a lack of public engagement and education, made it politically untenable. In the years that followed, however, San Diego’s dependence on increasingly strained imported water supplies, along with escalating costs and intensifying drought conditions across California, prompted a renewed urgency to find sustainable, local water sources (*Colorado region conditions*, n.d.).

Community involvement played a central role in reshaping the narrative around water reuse (Water reuse, 2012). Transparent communication, including accessible explanations of the advanced purification process and its proven effectiveness, was critical in countering earlier fears.

Nevertheless, the project was not without controversy. Critics voiced concerns over the program’s high financial cost (Schwartz, J, 2015). Others raised questions about the environmental impacts of increased energy consumption associated with advanced water treatment technologies (Schwartz, J, 2015). Still, the city’s leadership remained committed, strategically aligning Pure Water San Diego with state-level initiatives promoting water reuse.

Through careful planning, stakeholder engagement, and adaptive policymaking, San Diego ultimately succeeded where it once failed. The Pure Water program now serves as a national model for building public confidence in innovative water reuse solutions.

Planning and Implementation

Formal planning for Pure Water San Diego began in the early 2010s, emerging as a strategic response to the city's increasing vulnerability to water shortages and the imperative to develop a more sustainable, locally controlled water supply (City of San Diego, 2023).

Recognizing the long lead time required for complex infrastructure projects, city planners established a long-term implementation horizon, with the goal of meeting one-half of San Diego's potable water needs through purification of recycled water by 2035 (*City of San Diego official website*, n.d.a). This ambitious target laid the foundation for a phased rollout of the program, allowing for careful alignment with budget cycles, regulatory requirements, and evolving public sentiment (*City of San Diego official website*, n.d.a).

Phase 1 of Pure Water San Diego officially broke ground in 2019 and features the construction of the North City Pure Water Facility (*City of San Diego official website*, n.d.b). This facility is a central component of the program and integrates a multi-barrier advanced treatment process that includes microfiltration, reverse osmosis, and advanced oxidation (*City of San Diego official website*, n.d.b). The project also involves expanding existing wastewater collection infrastructure and constructing new pipelines to transport purified water to nearby reservoirs for blending before distribution (City of San Diego, 2023, *Diversification and*

operation, 2020). Future phases will expand treatment capacity and extend the system to other parts of the city (*City of San Diego official website*, n.d.a, *City of San Diego official website*, n.d.b).

Environmental review under the California Environmental Quality Act (CEQA) played a critical role in shaping the project's design and mitigation strategies (*CEQA: The California Environmental Quality act*, n.d.). The CEQA process required extensive assessment of the project's potential environmental impacts, including those related to air quality, traffic, noise, greenhouse gas emissions, biological resources, and marine ecosystems (*CEQA: The California Environmental Quality act*, n.d.). Based on these evaluations, the city adopted mitigation measures and design modifications to minimize negative impacts wherever feasible.

Stakeholder engagement has remained central throughout the planning and implementation phases. Advisory panels composed of representatives from environmental groups, engineering experts, public health officials, and community leaders provided input on technical and policy decisions (*SDCWA*, 2015). Public meetings, workshops, and formal comment periods offered residents multiple avenues to voice concerns, ask questions, and stay informed (*SDCWA*, 2015). This collaborative approach helped maintain public trust and transparency in the face of the project's scale and cost.

Managing the logistical and financial complexity of Pure Water San Diego has been one of its greatest challenges. Construction disruptions, interagency coordination, and escalating costs have all required continuous project management and communication (City of San Diego,

2023). To enhance public confidence, the city implemented a transparent budget tracking process and pursued state and federal funding, including low-interest loans through the EPA's Water Infrastructure Finance and Innovation Act (WIFIA) (Environmental Protection Agency, n.d.).

Environmental tradeoffs, particularly the increased energy demands of advanced water treatment, have also raised important sustainability considerations (Water Reuse, 2012).

Complementary initiatives, including regional water conservation programs, stormwater capture, and graywater reuse, further bolster the program's long-term sustainability goals.

Implications

The Pure Water San Diego project offers substantial long-term benefits, positioning the city as a leader in innovative and climate-resilient water management. By creating a reliable, locally controlled source of potable water, the program significantly reduces San Diego's dependence on imported supplies from the Colorado River and the California State Water Project, both of which are increasingly strained due to prolonged drought, climate change, and interstate competition (*Colorado region conditions*, n.d.). This strategic shift toward water independence enhances the city's resilience in the face of future water supply uncertainties.

Environmentally, one of the most immediate and tangible benefits of the project is the reduction in treated wastewater discharged into the Pacific Ocean (Water Reuse, 2012). By capturing this water and treating it to drinking water standards, the city not only preserves valuable resources but also helps protect marine ecosystems from nutrient loading and other

pollutants associated with traditional wastewater outfalls (Water Reuse, 2012). The program supports broader regional goals of integrated water management and ocean health.

Socially, Pure Water San Diego has played a role in shifting public perception around potable reuse. The once widespread stigma associated with "toilet to tap" has softened due to the city's transparent communication, facility tours, and community engagement strategies (Schwartz, J, 2015). However, public trust must be continually nurtured, especially among historically marginalized communities who may have lower trust in public utilities or feel excluded from decision-making processes. Ensuring that all residents, regardless of race, income, or neighborhood, are included in outreach and education efforts is essential for equitable implementation.

Economically, the project requires substantial upfront investment, with projected costs exceeding \$5 billion across its multiple phases (*City of San Diego official website*, n.d.b). While this may raise concerns among ratepayers, particularly those in lower-income brackets, long-term projections indicate that producing water locally will be more cost-effective than purchasing increasingly expensive imported supplies (City of San Diego, 2023). The project has also created local jobs in construction, engineering, and operations, stimulating the regional economy (City of San Diego, 2023). Moreover, by securing a more stable water supply, the program supports economic resilience and growth for sectors such as manufacturing, biotechnology, and tourism.

From a sustainability perspective, Pure Water San Diego represents a comprehensive model of integrated water resource management. The program aligns with the "One Water" approach, which treats all water, whether stormwater, wastewater, or drinking water, as part of a single, interconnected system (*City of San Diego official website*, n.d.a). By maximizing resource recovery and reducing waste, the project exemplifies circular economy principles in urban water planning. It also contributes to greenhouse gas reduction goals by reducing the need for long-distance water transport, though this benefit must be weighed against the high energy demands of advanced treatment processes (Water Reuse, 2012).

To enhance long-term sustainability, the city must continue investing in complementary initiatives such as energy efficiency, renewable energy procurement, and decentralized water strategies, such as stormwater harvesting and graywater reuse. Pure Water San Diego, therefore, stands not just as a technical achievement, but as a case study in the evolving relationship between cities, communities, and the water systems they depend on.

Recommendations

To ensure the long-term success and equity of the Pure Water San Diego project, it is essential to consider not only the technical and operational aspects, but also the social, environmental, and governance factors that influence public trust and system sustainability. The following recommendations aim to strengthen the project's benefits while addressing potential limitations. These strategies focus on improving community engagement, reducing the project's carbon footprint, encouraging complementary conservation efforts, and enhancing the flexibility

and accountability of water governance. Together, they offer a pathway for San Diego to lead in integrated, inclusive, and climate-resilient urban water management.

1. Enhance Equity-Focused Outreach

While the Pure Water San Diego project has made strides in public education and engagement, there remains a need to deepen outreach in historically underserved and marginalized communities. These communities may lack access to the same levels of information, trust in public agencies, or participation in water governance processes. To address this, the city should partner with trusted community-based organizations, schools, and local media to deliver multilingual, culturally appropriate educational materials about the safety, benefits, and necessity of potable reuse (Water Reuse, 2012). Hosting town halls in diverse neighborhoods, offering transportation to facility tours, and creating paid community ambassador programs can help bridge gaps in awareness and trust. This targeted outreach is not only a matter of social equity, but a critical element of long-term public buy-in and resilience planning.

2. Invest in Renewable Energy

Advanced water purification technologies such as reverse osmosis and advanced oxidation require substantial energy input, which can increase greenhouse gas emissions if sourced from fossil fuels (Water Reuse, 2012). To align the Pure Water program with broader sustainability goals, the city should invest heavily in renewable energy infrastructure to power its facilities. This can include onsite solar arrays, power purchase agreements for wind or solar energy, and

energy storage systems to manage peak demand (Water Reuse, 2012). Additionally, integrating energy-efficient design into facility upgrades and retrofits will further reduce the project's carbon footprint (Water Reuse, 2012). Over time, these investments can provide operational cost savings and demonstrate San Diego's leadership in climate-smart infrastructure.

3. Promote Complementary Conservation Programs

Potable reuse is only one piece of the puzzle for achieving long-term water sustainability. The city should complement Pure Water San Diego with aggressive promotion of conservation programs for households, businesses, and institutions. This includes rebates for installing greywater systems, rain barrels, drought-tolerant landscaping, and high-efficiency appliances (Water Reuse, 2012). Such efforts can reduce overall water demand, extend the benefits of Pure Water investments, and foster a culture of water stewardship (Water Reuse, 2012). By layering conservation incentives on top of the Pure Water program, the city can maximize its water security without relying solely on large-scale infrastructure.

4. Develop Adaptive Governance Structures

The success of Pure Water San Diego will depend not just on engineering and financing, but on the strength of its governance framework. As the project evolves, the city should adopt adaptive governance practices that allow for flexible, transparent decision-making. This includes establishing advisory boards with diverse community representation, creating mechanisms for regular performance reviews, and building in policy feedback loops based on monitoring data

and stakeholder input. Adaptive governance can also help the city respond to changing environmental conditions, regulatory updates, or shifts in public opinion. By institutionalizing transparency and participation, San Diego can build durable support for Pure Water while ensuring that it remains responsive to future needs.

Conclusion

Pure Water San Diego represents a landmark urban water initiative that redefines how modern cities can sustainably manage limited and increasingly uncertain water resources. Located in a semi-arid region and historically dependent on imported water from the Colorado River and Northern California, San Diego has long faced challenges related to water scarcity, price volatility, and environmental vulnerability. Pure Water San Diego tackles these issues head-on by transforming municipal wastewater into a safe, high-quality drinking water source using a multi-barrier treatment process that includes microfiltration, reverse osmosis, and advanced oxidation (*City of San Diego official website*, n.d.b). The initiative is projected to meet up to one-half of the city's future drinking water demand by 2035, marking a substantial step toward self-sufficiency (*City of San Diego official website*, n.d.a).

The project's significance extends beyond technical innovation. It exemplifies how cities can integrate infrastructure planning with robust public engagement, policy coordination, and environmental stewardship. Years of persistent outreach and transparency helped shift public perception from skepticism, rooted in failed efforts during the 1990s, to broad acceptance and

even support for potable reuse. Additionally, the city's emphasis on meeting and exceeding stringent regulatory standards has reinforced public trust and regulatory credibility.

However, challenges remain. The project involves substantial capital and operational costs and requires large amounts of energy, which raise concerns about greenhouse gas emissions and long-term affordability. Socially, it is critical that the benefits of the program, including improved water security and environmental health, are equitably shared across San Diego's diverse communities.

Despite these challenges, Pure Water San Diego stands as a compelling case study in urban resilience. It demonstrates how a city can adapt to climate pressures through science-based innovation, inclusive governance, and forward-thinking investment. Ongoing evaluations, adaptive management, and equity-focused policies will be essential to ensuring its long-term sustainability, but the project already offers a valuable model for other water-stressed cities around the world.

Sources

SDCWA. San Diego County Water Authority. (2025, April 1). <https://www.sdcwa.org/>

CEQA: The California Environmental Quality act. CEQA: The California Environmental Quality Act - Office of Land Use and Climate Innovation. (n.d.). <https://lci.ca.gov/ceqa/getting-started/>

City of San Diego official website. (n.d.a). Pure Water San Diego | City of San Diego Official Website. <https://www.sandiego.gov/public-utilities/sustainability/pure-water-sd>

City of San Diego official website. (n.d.b). Pure Water San Diego Program | City of San Diego Official Website. https://www.sandiego.gov/sites/default/files/2025-02/pw-prog-art-fact-sheet-021825_v5_0.pdf

City of San Diego official website. Pure Water San Diego | City of San Diego Official Website. (2024). <https://www.sandiego.gov/public-utilities/sustainability/pure-water-sd>

City of San Diego. “Pure Water San Diego – Phase 1 Construction Update.” YouTube, 15 Sept. 2023. <https://www.youtube.com/watch?v=RCUIZYOYZfA0&t=1s>

Colorado region conditions. Drought.gov. (n.d.). <https://www.drought.gov/watersheds/colorado>

Diversification and operation. San Diego County Water Authority | 2020 Annual Report. (2020.). <https://www.sdcwa.org/annualreport/2020/diversification-and-operation/>

Environmental Protection Agency. (2025a). *Water Quality Standards: Regulations and Resources*. EPA. <https://www.epa.gov/wqs-tech/what-are-water-quality-standards>

Environmental Protection Agency. (2025b). *Water Quality Standards: Regulations and Resources*. EPA. <https://www.epa.gov/wqs-tech/state-specific-water-quality-standards-effective-under-clean-water-act-cwa>

Environmental Protection Agency. (2024). *Water Quality Standards: Regulations and Resources*. EPA. <https://www.epa.gov/wqs-tech/relationship-between-water-quality-criteria-and-water-quality-standards>

Environmental Protection Agency. (n.d.). *Water Infrastructure Finance and Innovation Act (WIFIA)*. EPA. <https://www.epa.gov/wifia>

San Diego metro area population 1950-2025. MacroTrends. (n.d.). <https://www.macrotrends.net/global-metrics/cities/23129/san-diego/population>

Schwartz, J. (2015, May 8). *Water flowing from toilet to tap may be hard to swallow* - *The New York Times*. The New York Times.

<https://www.nytimes.com/2015/05/12/science/recycled-drinking-water-getting-past-the-yuck-factor.html>

Water reuse. (2012). *The National Academies*. <https://doi.org/10.17226/13303>